Serial No: 043921

Registration No: CIIT/SP13-BSB-024/ISB

By the Authority of the Board of Governors

COMSATS Institute of Information Technology

SANA ABDUL KHALIQ d/o ABDUL KHALIQ

Of

Islamabad Campus

has been conferred upon the degree of

Bachelor of Science in Bioinformatics

Given on this Fifteenth day of March two thousand and Seventeen at Islamabad

Date of Issuance: 15th March 2017

Controller of Examinations

Rector

Registrar

제 2024-0001848 호

학 위 증 명

과 정	: 일반대학원(M) 석사학위과정
학과 / 전공	: 융합의과학과(M) 의생명과학전공(M)
학 번	: 201845511
성 명	: KHALIQ SANA ABDUL
생 년 월 일	: 1994.12.16
학 위 명	: 이학 석사
학위 수여일	: 2020.08.27
학위 등록 번호	: 가천대19(석)618

위의 사실을 증명합니다.

2024년 1월 4일

가 천 대 학 교 총 장

[인터넷 발급]우리 대학 및 www.certpia.com 원본대조에뉴에서 인터넷 발급번호로 입력(발급일로부터 90일이내) 또는 스마트폰 원본대조앱으로 증명서 하단의 QR코드를 촬영하여 원본대조 확인 가능. 스마트 원본대조앱은 앱스토어/마켓에서 원본대조로 검색 다운받을수 있습니다.

위·변조방지용
원본대조코드
특허10-1039390

(Internet no) K01157294404

GACHON UNIVERSITY

1342, Seongnam-daero, Sujeong-gu, Seongnam-si, Gyeonggi-do, Republic of Korea, 13120.

Graduate School

CERTIFICATE OF GRADUATION

Date of Issuance : Feb. 28, 2024

No. : 2024-0031

Name in Full	:	SANA ABDUL KHALIQ
Student I.D.No.	:	202055511
Sex	:	Female
Date of Birth	:	Dec. 16, 1994
Nationality	:	Pakistan
Date of Admission	:	Sep. 01, 2020
Department	:	Health Sciences and Technology
Major Field	:	Bio-Medical Science (Ph. D. Course)
Degree Received	:	Doctor of Philosophy in Science
Date of Graduation	:	Feb. 22, 2024

This is to certify that the above is true.

Seal

Min, Se Hong

Dean, Graduate School

Gachon University

GACHON UNIVERSITY

1342 Seongnamdaero, Sujeong-gu, Seongnam-si, Gyeonggi-do, Korea, 461-701

CERTIFICATE OF EMPLOYMENT

Date of Issuance : Sep. 15, 2025

No. : 2025-499

Name in Full : Khaliq Sana Abdul

Date of Birth : Dec. 16, 1994 Sex : Female

Nationality : Pakistan

Department	Period	Position
College of Medicine Premedical Course	Mar. 01, 2024 ~ Present	Assistant Professor

This is to certify that the above mentioned fact is true and correct.

Seal

GIL YA LEE M.D., Ph.D.
President
Gachon University

Karolinska Institutet

SCHOLARSHIP
2025-09-10

Surname	First name	Swedish national registration number
Khaliq	Sana Abdul	19941216-K2KK
Street address 16,Hambak-ro,35 Beon-Gil,Yeonsu-gu,Incheon, South Korea		Telephone no. +821058998781
Postcode and town 21912 Yeonsu-gu		Journal no.
Department		

This scholarship regards

Undergraduate studies ☐	Doctoral studies (complementary) ☐	Postdoc studies ☒ Decisions can only be made when the person in question has completed their dissertation/equivalent

Date (ÅÅMMDD - ÅÅMMDD)	The amount is regulated by KI:s guidelines, for now: 171000 SEK semiannually
251201 - 261130	

Percentage	Project	Percentage	Project
100	K26200593		

For doctoral students only

Name of external grant awarding body
Amount paid by external awarding body

Signature:

The decision has been taken according to KI's scholarship regulation ☒
The scholarship holder has not received a salary or fees from KI within a two year period prior to the first scholarship payment, with the exeption of smaller amounts according KI's scholarship regulation ☒

10/9-25
Date and signature head of administration

Helene Utterberg
Administrativ chef/
Head of Administration

Se digital signn / Helene
Date and signature immediate superior (manager)

Approval of scholarship and payment decision

2025-09-10
Date and signature head of department

Marie Wahren Herlenius
Head of Department
Prefekt

Two master copies. One for the department's archiving together with enclosures according to Checklista och ansvar i stipendieärenden (only in Swedish). One for the HR-department/salary unit.

KSCB 23-5608

THE KOREAN SOCIETY FOR CELL BIOLOGY

Best Poster Award
to

Sana Abdul Khaliq
Gachon University

"ARHGEF3 Enhances Adipogenesis through the PPARγ and YAP-RhoA Pathways in a Manner Dependent on its GEF Activity.

at the 56th KSCB Conference, Daejeon, Korea
Aug 23, 2023

The President of the Korean Society for Cell Biology
Do Sik Min, Ph.D.

GENOLUTION

Excellent Poster Awards

Sana Abdul khaliq

Gachon University South Korea

for the outstanding poster presentation entitled

C-peptide Selectively Enhances Migration in Human Endometrial Stromal Cells by Controlling the Expression of MMP9 and TIMP1/3 Through Direct Regulation of β-catenin in Akt and PP1 Dependent Manner

at the International Conference of the Korean Society for
Molecular and Cellular Biology (KSMCB)
November 3 - November 5, 2021

Rho Hyun Seong

Rho Hyun Seong, Ph.D.
President, KSMCB

Excellent Poster Awards

Sana Abdul khaliq

Gachon University

for the outstanding poster presentation entitled

C-peptide Inhibits Decidualization in Human Endometrial Stromal Cells via GSK3β-PP1

at the International Conference of the Korean Society for
Molecular and Cellular Biology (KSMCB)
October 5 - October 7, 2020

Jin Won Cho, Ph.D.
President, KSMCB

특허증

CERTIFICATE OF PATENT

특 허 Patent Number	제 10-2640230 호
출원번호 Application Number	제 10-2021-0126090 호
출원일 Filing Date	2021년 09월 24일
등록일 Registration Date	2024년 02월 20일

발명의명칭 Title of the Invention

C-peptide를 처리한 자궁내막 기질세포의 이동성 변화 기작을 이용한 자궁내막 관련질환의 진단 방법 및 자궁내막 관련질환 치료제의 스크리닝 방법

특허권자 Patentee

가천대학교 산학협력단(131171-*****)**
경기도 성남시 수정구 성남대로 1342 (복정동)

발명자 Inventor

등록사항란에 기재

위의 발명은 「특허법」에 따라 특허원부에 등록되었음을 증명합니다.

This is to certify that, in accordance with the Patent Act, a patent for the invention has been registered at the Korean Intellectual Property Office.

2024년 02월 20일

QR코드로 현재기준
등록사항을 확인하세요

특허청장
COMMISSIONER,
KOREAN INTELLECTUAL PROPERTY OFFICE

특허청
Korean Intellectual Property Office

등록사항

특 허 **제 10-2640230 호**
Patent Number

발명자 Inventors

윤미섭(671016-*****)**
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Gmail

Sana Abdul Khaliq <sanaakhaliq1@gmail.com>

Outcome of IUBMB Travel Fellowship application

4 messages

James Murphy <jamesm@wehi.edu.au> Wed, Apr 3, 2024 at 12:27 PM
To: "sanaakhaliq1@gmail.com" <sanaakhaliq1@gmail.com>

Dear Sana

Many thanks for your application for an IUBMB Travel Fellowship to support your travel to the upcoming Biomolecular Horizons 2024 congress in Melbourne in September this year. Firstly, apologies for the delay in notifying you of the outcome of your application. We were overwhelmed with the number of applications following extension of the deadline, and also had to navigate the assessment process around the Easter holiday period in Australia.

I am delighted to advise that you have been successful in your application. You have been awarded **$USD500** to support your return travel to Melbourne. If you have applied for the Young Scientists Program, please note that these applications are currently being evaluated separately. Should your Young Scientists Program application be successful, you will not be awarded both IUBMB Fellowship and YSP funds; IUBMB Fellowship support will instead be transferred to a waitlisted applicant. Please also note that the funds awarded to you will be reimbursed following attendance at Biomolecular Horizons 2024. We will be in touch to make arrangements to transfer the funds in due course.

On behalf of the assessment committee and the IUBMB, please accept my congratulations on your award.

Best wishes

James

--
James M. Murphy, PhD (*he/him*)

Head, Inflammation Division

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ARHGEF3 coordinates adipocyte hypertrophy and differentiation through dual YAP-RhoA and PPARγ activation

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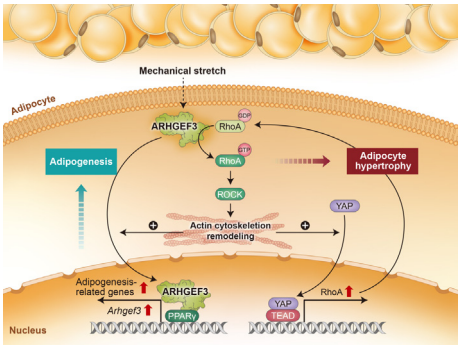
[j] Lee Gil Ya Cancer and Diabetes Institute, Gachon University, Incheon 21999, Republic of Korea

HIGHLIGHTS

- ARHGEF3 regulates adipogenesis through RhoA-YAP signaling for hypertrophy while promoting differentiation via PPARγ activation.
- ARHGEF3 deletion reduces fat accumulation by decreasing RhoA, impairing cytoskeletal reorganization and limiting adipocyte expansion.
- ARHGEF3 enables YAP nuclear transport to bind TEAD motifs on RhoA promoter, creating a feedback loop for cytoskeletal remodeling.
- ARHGEF3 and PPARγ form a regulatory circuit: ARHGEF3 enhances PPARγ activity while PPARγ upregulates ARHGEF3, boosting adipogenesis.

GRAPHICAL ABSTRACT



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ABSTRACT

Introduction: Obesity presents a significant global health burden, necessitating insights into the molecular drivers of adipogenesis and adipose tissue regulation.
Objectives: This study investigates the role of Rho guanine nucleotide exchange factor 3 (ARHGEF3) in adipocyte differentiation and hypertrophy, focusing on its influence on adipogenesis and body weight regulation under high-fat diet conditions.
Methods: ARHGEF3$^{-/-}$ mice and littermate controls were subjected to a high-fat diet (HFD) and underwent comprehensive metabolic phenotyping. In vitro studies in C3H10T1/2 cells were conducted to assess ARHGEF3's role in adipogenesis, utilizing quantitative PCR, western blotting, chromatin

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